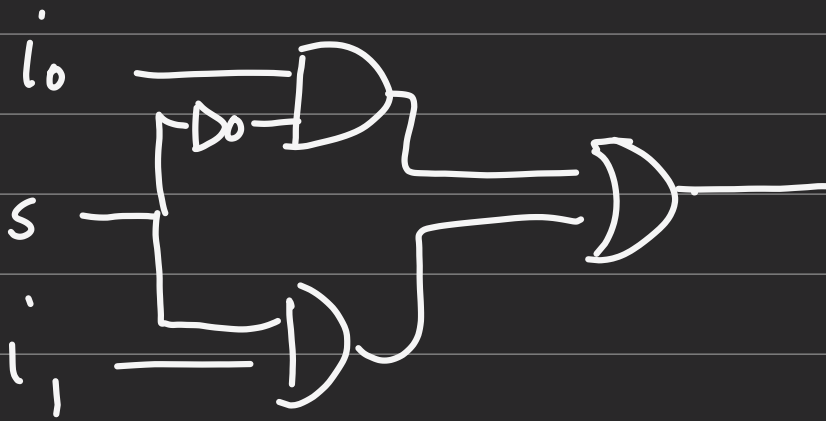


Hierarchical Design

① 2X1 Mux

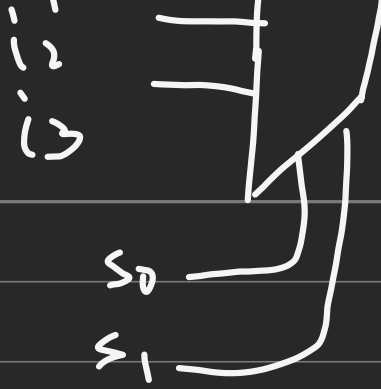


$$o = \sim s \& i_0 \mid s \& i_1$$



② 4X1 Mux





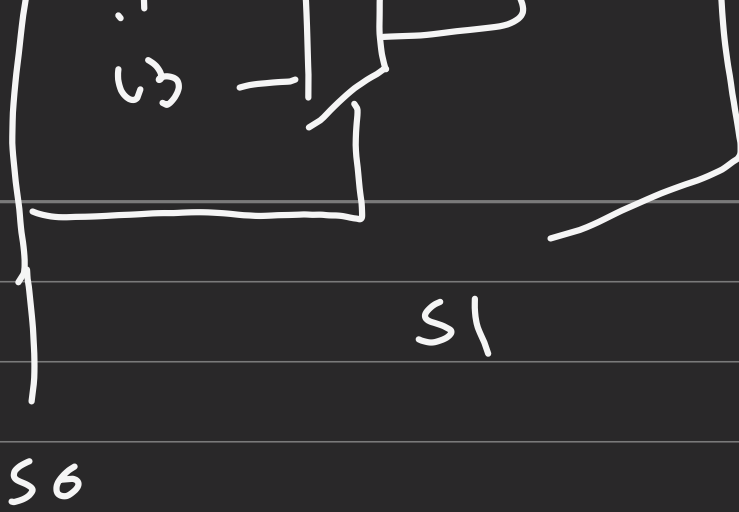
$$0 = \sim s_0 \& \sim s_1 \& i_0 \quad \text{---}$$

$$\text{---} \quad | s_0 \& s_1 \& i_3$$

↳ using 2x1 Mux

s_0	s_1	o
0	0	i_0
0	1	i_1
1	0	i_2
1	1	i_3





↳ Creating instance

ie

```

mem 2x1 M0(x0, x1, s0, g0);
mem 2x1 M1(x2, x3, s0, g1);
mem 2x1 M2(g0, g1, s1, f);

```

OR

```

mem 2x1 M2(.x1(g0), .x2(g1)
            .s(s1), .f(f));

```

* order of these instances doesn't matter
since we are not using always block